

Comparison of Lightweight Encryption Algorithms

B.Tech Final Year Project

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Outline

- 1 Introduction
- 2 Experimental Methodology
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Historical Context

- DES (1977) → AES (2001): Evolved for desktop/server computing
- IoT devices: constrained CPU, memory, power — traditional algorithms impractical
- Mirai botnet (2016): Exploited insecure embedded devices — millions hijacked

Problem

- Developers deploy weakened or ad hoc security
- Lightweight cryptography: efficient primitives for constrained platforms
- Need to balance security, performance, and implementation cost

Objectives & Contributions

This study aims to:

- Evaluate PRESENT, SPECK, and ASCON on Raspberry Pi Pico (ARM Cortex-M0+)
- Compare execution time, memory usage, and throughput
- Identify design trade-offs for constrained embedded environments

Contributions:

- Reproducible Rust-based benchmarking framework for RP2040
- Empirical data across block cipher, mode, and payload-size dimensions
- Practical deployment recommendations based on measured trade-offs

Property	PRESENT	SPECK	ASCON
Block Size	64 bits	64 bits	64 bits
Key Sizes	80, 128 bits	64–256 bits	128 bits
Structure	SPN	ARX Feistel	Sponge-Duplex
Rounds	31/32	22-34	12
Standardization	ISO 29192-2	—	NIST SP 800-232
AEAD Support	No	No	Yes

Key Differentiators:

- **PRESENT**: ~1,570 GE (hardware-optimized)
- **SPECK**: 42 CPB (software fastest)
- **ASCON**: AEAD + hashing built-in

Hardware: Raspberry Pi Pico (ARM Cortex-M0+, 133 MHz, 264 KB SRAM)

Metrics: Throughput (Mbps), Cycles/Byte, ROM footprint, RAM usage

Procedure: 1000+ iterations, hardware timer, test vector validation

Block Cipher Benchmarks

Algorithm	CPB	Mbps
PRESENT-80	2,958	0.34
PRESENT-128	3,068	0.33
SPECK-64/96	42	23.81
SPECK-64/128	42	23.67
ASCON-128	132	7.56

Key Finding: SPECK is **70x faster** than PRESENT and **3x faster** than ASCON.

Why This Performance Gap?

Architectural Explanation:

- **SPECK (ARX)**: Add, Rotate, XOR $\rightarrow$ single-cycle ARM instructions (ADDS, RORS, EORS) — no memory lookups
- **PRESENT (SPN)**: Bit-permutation requires 64 separate bit extractions/reinsertions per round — 31 rounds (PRESENT-80) / 32 rounds (PRESENT-128) = 70x overhead
- **ASCON (Sponge)**: 64-bit words align with Cortex-M0+ datapath, but 12-round p^a permutation is heavier than SPECK's 5-instruction round

Bottom Line: Hardware-optimized ciphers (PRESENT) sacrifice software performance; software-optimized ciphers (SPECK) exploit architectural fit.

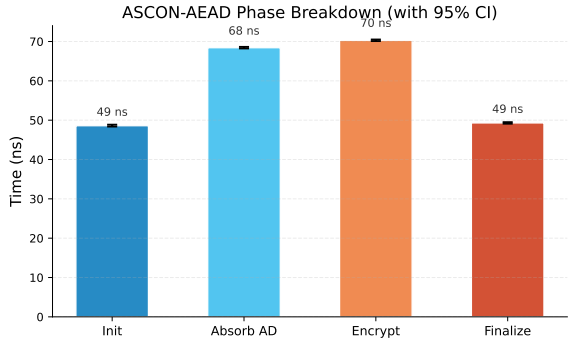
Memory Footprint

Cipher	Key	State	ROM	RAM
PRESENT-80	80 bits	8 bytes	~2 KB	~16 B
PRESENT-128	128 bits	8 bytes	~2.5 KB	~16 B
SPECK-64/96	96 bits	16 bytes	~1 KB	~32 B
SPECK-64/128	128 bits	16 bytes	~1.2 KB	~32 B
ASCON-128	128 bits	40 bytes	~3 KB	~40 B

SPECK: Smallest ROM (~1 KB) — ARX-only ops, no lookup tables

ASCON Phase Breakdown

Phase	Cycles
Initialization	427
Absorb (AD)	216
Encrypt	464
Finalize	422
Total	1,529



Cipher	16B	64B	256B	1024B
SPECK-64/128 (CTR)	0.29	0.27	0.26	0.25
ASCON-128 (AEAD)	2.56	0.64	0.51	0.47
PRESENT-128 (CTR)	27.18	27.07	27.17	27.00

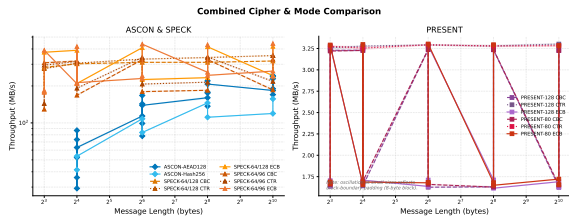
- SPECK: 108x more energy-efficient than PRESENT
- ASCON: $\sim 41 \mu\text{J}$ initialization overhead (constant), but per-byte cost scales from $2.56 \rightarrow 0.47 \mu\text{J}$ as payload grows
- Ideal for battery-powered IoT sensors

Speedup Analysis

Performance relative to ASCON:

Cipher	64B	256B	1024B
SPECK (ECB)	1.34x	1.08x	1.03x
PRESENT	0.012x	0.010x	0.009x

Insight: ASCON's 427-cycle init overhead amortizes at larger payloads; SPECK advantage shrinks from 34% to 3%.



Scorecard Comparison

Metric	PRESENT	SPECK	ASCON
Throughput (Mbps)	0.34	23.67	7.56
CPB (cycles)	2,958	42	132
ROM (KB)	2.5	1.2	3.0
RAM (bytes)	16	32	40
AEAD	No	No	Yes
NIST Standard	No	No	Yes
ISO Standard	Yes	No	No

- **ASCON**: NIST SP 800-232, proven indistinguishability bounds, no attacks on full 12-round permutation
- **PRESENT**: ISO/IEC 29192-2, provable SPN bounds (max diff probability 2^{-2}), confidentiality only
- **SPECK**: No provable bounds, best attacks reduce 20/27 rounds; ISO rejection (NSA origin, geopolitical objections)
- **Side-channels**: SPECK vulnerable to CPA (50 traces on 8-bit); PRESENT's constant-time ops more resistant

- ➊ **Real-time sensor data:** SPECK-64/128 (CTR) — 23.67 Mbps, 32B RAM
- ➋ **Secure IoT with auth:** ASCON — 7.56 Mbps, AEAD, NIST standardized
- ➌ **RFID tags:** PRESENT — ultra-low gate count (1570 GE)
- ➍ **Firmware authentication:** ASCON-hash — 320-bit state, 12-round p^a
- ➎ **Hybrid deployment:** SPECK (bulk) + ASCON (auth tags)
- ➏ **Legacy integration:** PRESENT-128 (CBC) — backward compatibility

- **Side-channels:** Not analyzed; SPECK shows vulnerability to CPA attacks
- **Scope:** Single platform (RP2040), software-only implementations
- **Future:** Hardware-accelerated implementations, side-channel countermeasures

Thank You

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